

TOCOL Function in Excel

1. What is TOCOL?

The TOCOL function takes a range with many rows and columns and stacks all its values into a single column. It solves the problem of data that is spread across a grid when you need one simple list. A Data Analyst uses it to flatten tables so the values can be sorted, counted, filtered, or passed to other functions.

2. Syntax

`=TOCOL(array, [ignore], [scan_by_column])`

Argument	Required?	What it accepts
array	Required	The range or array to convert into one column (for example A1:C3)
ignore	Optional	What to skip: 0 keep all (default), 1 ignore blanks, 2 ignore errors, 3 ignore blanks and errors
scan_by_column	Optional	FALSE (default) reads row by row, left to right. TRUE reads column by column, top to bottom.

To use `scan_by_column` and keep the default for `ignore`, leave the middle argument empty or use 0: `=TOCOL(A1:C3, , TRUE)`.

3. Simple Example

Turn a 3 × 3 grid into one column, ignoring blank cells:

	A	B	C	E
1	Apple	Banana	Orange	Apple
2	Mango		Grape	Banana
3	Kiwi	Pineapple		Orange
4				Mango
5				Grape
6				Kiwi
7				Pineapple

Formula in E1 only: `=TOCOL(A1:C3, 1)`

Excel reads row 1 (Apple, Banana, Orange), then row 2, then row 3, skipping the two blank cells (B2 and C3). The 7 values spill down E1 to E7. **Result: 7 values in one column**

4. Another Example

Stack monthly sales into one column, reading column by column:

	A	B	C	D	F
1	Region	Jan	Feb	Mar	Result
2	North	500	700	650	500
3	South	400	550	600	400
4					700
5					550
6					650
7					600

Formula in F2 only: `=TOCOL(B2:D3, , TRUE)`

With TRUE, Excel reads down the Jan column first (500, 400), then Feb (700, 550), then Mar (650, 600). With the default FALSE it would read row by row: 500, 700, 650, 400, 550, 600.

5. When to Use It

- Flatten a multi-column table into one list for analysis
- Combine several columns of names, IDs, or codes into one list
- Remove blanks and errors while reshaping data
- Prepare a list for functions such as UNIQUE, SORT, or COUNTIF
- Prepare wide data for charts or pivot tables

6. Important Things to Remember

- TOCOL is a dynamic array function. You type one formula and the results spill down into the cells below.
- By default, TOCOL reads row by row (`scan_by_column` is FALSE). Set it to TRUE to read column by column.
- By default, nothing is ignored, so blank cells appear in the result as 0 and errors are kept.
- Use 1 to skip blanks, 2 to skip errors, and 3 to skip both.
- The cells below the formula must be empty, or Excel shows #SPILL!
- TOCOL is available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier.

7. Common Mistakes

Mistake: Expecting blank cells to be skipped automatically, then finding 0 values in the result. **Fix:** Add the ignore argument: `=TOCOL(A1:C3, 1)`.

Mistake: Using 1 to remove blanks and errors, but still seeing #N/A. **Fix:** Use 3 to ignore both blanks and errors, or 2 for errors only.

Mistake: Getting the wrong order because of the reading direction. **Fix:** Remember that row by row is the default. Add TRUE as the third argument for column by column.

Mistake: Getting #SPILL! because cells below the formula already contain data. **Fix:** Clear those cells or move the formula to an empty area.

8. Data Analyst Tip

Combine TOCOL with other functions to build clean lists. For example, `=SORT(UNIQUE(TOCOL(A1:C10, 3)))` flattens a table, removes blanks and errors, removes duplicates, and sorts the result in one formula.